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POINCARÉ & LOG-SOBOLEV FOR THE ORNSTEIN–UHLENBECK AND LANGEVIN–SCHMOLUCHOWSKI SEMIGROUPS

- $B(t),\ 0 \le t < \infty$: Standard Brownian Motion in $\mathbb{R}^n$
- γ : Standard Gaussian distribution on $\mathcal{B}(\mathbb{R}^n)$, i.e.,

$$\gamma(dy) = \frac{1}{(2\pi)^{n/2}} e^{-\frac{1}{2}|y|^2}\, dy$$

- $X(t),\ 0 \le t < \infty$: Ornstein–Uhlenbeck diffusion process

$$dX(t) = -X(t)\,dt + \sqrt{2}\, dB(t), \quad X(0) = \xi;$$

equivalently,

$$X(t) = e^{-t}\left(\xi + \sqrt{2}\int_0^t e^{s}\, dB(s)\right)$$

where ξ is a random variable independent of $B(\cdot)$.

- Then

$$\underline{(P_t f)(x)} \triangleq \mathbb{E}^x f(X(t))$$

$$\underline{= \int_{\mathbb{R}^n} f\left(xe^{-t} + y\sqrt{1-e^{-2t}}\right) \gamma(dy)}$$

defines the so-called <u>ORNSTEIN–UHLENBECK Semigroup</u> for $t \in [0,\infty)$, $f \in C_b(\mathbb{R}^n)$.

This has the following properties.

- $P_0 = I =$ identity
- $[0,\infty) \ni t \longrightarrow P_t f \in \mathbb{L}^2(\mathbb{R}^n, \gamma)$ is continuous, for every given $f \in C_b(\mathbb{R}^n)$
- $P_t(P_s f) = P_{t+s} f = P_s(P_t f)$, $\forall\, s \geq 0,\, t \geq 0$
- $P_t 1 \equiv 1$; $P_t f \geq 0$ if $f \geq 0$
- $\|P_t f\|_\infty \leq \|f\|_\infty$, $\forall\, t \geq 0,\, f \in C_b(\mathbb{R}^n)$.

PROPOSITION: The linear operator

(1) $$Lf \triangleq \Delta f - \langle x, \nabla f \rangle$$

is the infinitesimal generator of the OU-semigroup:

(2) $$\frac{\partial}{\partial t}(P_t f)(x) = L(P_t f)(x) = P_t(Lf)(x)$$

for every $t \geq 0$, $x \in \mathbb{R}^n$, $f \in C_b^2(\mathbb{R}^n)$. Also $(Lf)(x) = \lim_{t \downarrow 0} \frac{(P_t f)(x) - f(x)}{t}$.

PROPOSITION: The probability measure γ is the unique invariant measure for the OU-semigroup, i.e.,

(3) $$\int_{\mathbb{R}^n} (P_t f)(x)\, \gamma(dx) = \int_{\mathbb{R}^n} f(x)\, \gamma(dx), \quad \forall\, f \in C_b(\mathbb{R}^n),\ \forall\, t \geq 0$$

and is also ergodic:

(4) $$\lim_{t \to \infty} (P_t f)(x) = \int_{\mathbb{R}^n} f(y)\, \gamma(dy); \quad \forall\, x \in \mathbb{R}^n,\ f \in C_b(\mathbb{R}^n).$$

Remark: The invariance property (3) is tantamount to

(5)' $$\int_{\mathbb{R}^n} (Lf)(y) \cdot \gamma(dy) = 0, \quad \forall f \in C_b^2(\mathbb{R}^n).$$

In fact, we have the stronger property

(5) $$\underbrace{\int_{\mathbb{R}^n} g(y) \cdot (Lf)(y) \cdot \gamma(dy) = \int_{\mathbb{R}^n} f(y) \cdot (Lg)(y) \cdot \gamma(dy) = -\int_{\mathbb{R}^n} \langle \nabla f(y), \nabla g(y) \rangle \gamma(dy)}_{\mathcal{E}(f,g)}$$

for every f, g in $C_b^2(\mathbb{R}^n)$.

Proof of (5): This is a simple integration by parts, noting that the probability density

$$\varphi(y) = \frac{1}{(2\pi)^{n/2}} e^{-\frac{1}{2}|y|^2}$$

of the standard Gaussian measure γ, satisfies the property

(6) $$\boxed{\nabla \varphi(y) = -y\, \varphi(y),} \quad y \in \mathbb{R}^n.$$

Proof of (2): We have clearly that

$$\frac{\partial}{\partial t}(P_t f)(x) = \int_{\mathbb{R}^n} \left(-x e^{-t} + y \frac{e^{-2t}}{\sqrt{1-e^{-2t}}} \right) \nabla f\left(x e^{-t} + y\sqrt{1-e^{-2t}} \right) \gamma(dy)$$

is the sum of the two terms

(7) $$-x e^{-t} \int_{\mathbb{R}^n} \nabla f\left(x e^{-t} + y\sqrt{1-e^{-2t}} \right) \gamma(dy) = -x e^{-t} \left(P_t(\nabla f) \right)(x) = -x \cdot \nabla (P_t f)(x)$$

and

$$\frac{e^{-2t}}{\sqrt{1-e^{-2t}}} \int y \cdot \nabla f\left(x e^{-t} + y\sqrt{1-e^{-2t}} \right) \varphi(y)\, dy = \Delta (P_t f)(x).$$

For this last identity we integrate by parts, recalling (6).

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<u>Remark:</u> (i) The equality (5) is equivalent to (†)

$$(5)' \qquad \int_{\mathbb{R}^n} g \cdot P_t f \cdot dy = \int_{\mathbb{R}^n} f \cdot P_t g \cdot dy \,, \quad \forall\, t \geq 0.$$

Then with $X(0) \sim \mathcal{J}$ (starting the diffusion at its invariant distribution) and selecting

$$f = \mathbb{1}_A \,, \; g = \mathbb{1}_B$$

we obtain <u>reversibility</u>:

$$\mathbb{P}[X(t) \in A,\, X(0) \in B] = \mathbb{P}[X(0) \in A,\, X(t) \in B].$$

(ii) We introduce also the DIRICHLET Form

$$\mathcal{E}(f,g) \triangleq \int_{\mathbb{R}^n} \langle \nabla f, \nabla g \rangle \, dy = \int_{\mathbb{R}^n} f\,(-Lg)\, dy = \int_{\mathbb{R}^n} g\,(-Lf)\, dy$$

and the DIRICHLET Energy

$$\mathcal{E}(f,f) \triangleq \int_{\mathbb{R}} |\nabla f|^2 \, dy = \int_{\mathbb{R}} f\,(-Lf)\, dy \geq 0.$$

In particular, $-L$ is a positive operator.

(†) Because $(5)'$ amounts to $\int_{\mathbb{R}^n} g(P_t f - f)\, dy = \int_{\mathbb{R}^n} f(P_t g - g)\, dy$, and this leads directly to (5).

And conversely...

Remark: As we noticed in (7), $\nabla(P_t f)(x) = e^{-t} \cdot P_t(\nabla f)(x)$ holds for $t \geq 0$, $x \in \mathbb{R}^n$ and $f \in C_b^2(\mathbb{R}^n)$. In particular, we have also the commutation property

(8) $$\|\nabla(P_t f)(x)\| \leq e^{-t} \cdot P_t(\|\nabla f\|)(x),$$

for all norms $\|\cdot\|$ on $\mathbb{R}^n$.

We define now, for $f \in C_b^2(\mathbb{R}^n)$, its variance

(9) $$\mathrm{Var}_\gamma(f) \triangleq \int_{\mathbb{R}^n} f^2 d\gamma - \left(\int_{\mathbb{R}^n} f \, d\gamma\right)^2$$

relative to the standard Gaussian γ.

THEOREM: POINCARÉ INEQUALITY. The quantity (9) satisfies

(10) $$\mathrm{Var}_\gamma(f) \leq \int_{\mathbb{R}^n} |\nabla f|^2 d\gamma = \mathcal{E}(f,f)$$

with equality if $\nabla f = C \in \mathbb{R}^n$.

"Dirichlet Energy controls the Variance"

(10)′ Proof: We note $P_0 f = f$, $P_\infty f = \int_{\mathbb{R}^n} f \, d\gamma$, introduce $V^f(t) \triangleq \int_{\mathbb{R}^n} (P_t f)^2 d\gamma$,

(10)″ and observe $$\mathrm{Var}_\gamma(f) = V^f(0) - V^f(\infty) = -\int_0^\infty \frac{d}{dt} V^f(t) \, dt.$$

But from (5) we observe

$$\frac{d}{dt} V^f(t) = 2\int_{\mathbb{R}^n} \frac{d}{dt} P_t f \cdot P_t f \, d\gamma =$$

(10)‴ $$= 2\int_{\mathbb{R}^n} L(P_t f) \cdot P_t f \, d\gamma = -2\int_{\mathbb{R}^n} |\nabla(P_t f)|^2 d\gamma = -2\int_{\mathbb{R}^n} \mathcal{E}(P_t f, P_t f) \, d\gamma;$$

invoking now (8), we obtain

$$\mathrm{Var}_\gamma(f) = 2\int_0^\infty\left(\int_{\mathbb{R}^n}|\nabla(P_t f)|^2\,d\gamma\right)dt \le 2\int_0^\infty e^{-2t}\left(\int_{\mathbb{R}^n}\left(P_t(|\nabla f|)\right)^2 d\gamma\right)dt$$

$$\le 2\int_0^\infty e^{-2t}\left(\int_{\mathbb{R}^n}P_t(|\nabla f|^2)\,d\gamma\right)dt.$$

But the standard Gaussian distribution is invariant under the action of the semigroup $(P_t)_{t\ge 0}$, therefore the spatial integral of the last display is constant, and

$$\mathrm{Var}_\gamma(f) \le 2\int_0^\infty e^{-2t}\left(\int_{\mathbb{R}^n}|\nabla f|^2\,d\gamma\right)dt = \int_{\mathbb{R}^n}|\nabla f|^2\,d\gamma,$$

as claimed.

We note that all inequalities above, hold as equalities for functions f with constant gradient $\nabla f = C \in \mathbb{R}^n$. $\square$

For a given function $f : \mathbb{R}^n \to [0,\infty)$ with $\int_{\mathbb{R}^n} f\,d\gamma > 0$, we define now the <u>relative entropy</u>

(11) $$\mathrm{Ent}_\gamma(f) \triangleq \int_{\mathbb{R}^n} f \log\left(\frac{f}{\int_{\mathbb{R}^n} f\,d\gamma}\right)d\gamma = \mathbb{E}_\gamma[h(f)] - h(\mathbb{E}_\gamma(f)) \ge 0.$$

Here, $\mathbb{E}_\gamma(\cdot)$ denotes integration in $\mathbb{R}^n$ with respect to the standard Gaussian probability measure γ, and

(11)' $$h(x) \triangleq x.\log x, \quad x > 0$$

is a convex function.

<u>THEOREM</u>: <u>LOG-SOBOLEV INEQUALITY</u>. In the above context,

(12) $$\mathrm{Ent}_\gamma(f) \le \frac{1}{2}\int_{\mathbb{R}^n}\frac{|\nabla f|^2}{f}\,d\gamma = \frac{1}{2}I_\gamma(f).$$

And again, this holds as equality when $\nabla f = C$ is a constant in $\mathbb{R}^n$.

Remark: This inequality appears often with f replaced by f^2, in which case it reads

(12)′ $$\frac{1}{2} \operatorname{Ent}_\gamma (f^2) \leq \int_{\mathbb{R}^n} |\nabla f|^2 \, d\gamma = \mathcal{E}_\gamma(f,f) .$$

"Dirichlet Energy Controls the Relative Entropy"

Proof: We mimic the previous argument: we introduce $E^f(t) \triangleq \int_{\mathbb{R}^n} P_t f \cdot \log P_t f \, d\gamma$, recall $P_0 f = f$, $P_\infty f = \int_{\mathbb{R}^n} f \, d\gamma$, and note

$$\operatorname{Ent}_\gamma (f) = \int_{\mathbb{R}^n} f \log f \, d\gamma - \int_{\mathbb{R}^n} f \, d\gamma \cdot \log\left(\int_{\mathbb{R}} f \, d\gamma\right) = E^f(0) - E^f(\infty)$$
$$= -\int_0^\infty \frac{d}{dt} E^f(t) \, . \, dt .$$

Now from (2), (5′) and (5), we get

$$\frac{d}{dt} E^f(t) = \frac{d}{dt} \int_{\mathbb{R}^n} h(P_t f) \, . \, d\gamma = \int_{\mathbb{R}^n} h'(P_t f) \, . \, L(P_t f) \, . \, d\gamma$$
$$= \int_{\mathbb{R}^n} L(P_t f) \, . \, \log(P_t f) \, . \, d\gamma = -\int_{\mathbb{R}^n} \nabla(P_t f) \, . \, \nabla(\log P_t f) \, . \, d\gamma$$
$$= -\int_{\mathbb{R}^n} \frac{|\nabla(P_t f)|^2}{P_t f} \, d\gamma$$

where again $h(x) \triangleq x \, . \log x$.

The quantity

(13) $$\boxed{I_\gamma (f) \triangleq \int_{\mathbb{R}} \frac{|\nabla f|^2}{f} \, d\gamma} \quad \left(= \mathbb{E}^\mu\left(|\nabla \log f|^2\right), \ f = \frac{d\mu}{d\gamma} \right)$$

is the FISHER Information of f with respect to the standard GAUSSian measure γ.

(DE BRUJN)

We have derived the identity
$$\frac{d}{dt} E^f(t) = -I_\gamma(P_t f),$$
which gives
$$\mathrm{Ent}_\gamma(f) = E^f(0) - E^f(\infty) = \int_0^\infty I_\gamma(P_t f) = \int_0^\infty \left(\int_{\mathbb{R}^n} \frac{|\nabla (P_t f)|^2}{P_t f} d\gamma \right) dt$$
$$\leq \int_0^\infty e^{-2t} \left(\int_{\mathbb{R}^n} \frac{(P_t(|\nabla f|))^2}{P_t f} d\gamma \right) dt,$$
by virtue of (8).

But from CAUCHY-SCHWARZ we get
$$P_t(|\nabla f|) = P_t\left(\frac{|\nabla f|}{\sqrt{f}} \sqrt{f} \right) \leq \sqrt{P_t\left(\frac{|\nabla f|^2}{f} \right) \cdot P_t(f)},$$
thus
$$\frac{P_t(|\nabla f|)^2}{P_t(f)} \leq P_t\left(\frac{|\nabla f|^2}{f} \right).$$
Substituting, we get
$$\mathrm{Ent}_\gamma(f) \leq \int_0^\infty e^{-2t} \left(\int_{\mathbb{R}^n} P_t\left(\frac{|\nabla f|^2}{f} \right) d\gamma \right) dt = \frac{1}{2} \int_{\mathbb{R}^n} \frac{|\nabla f|^2}{f} d\gamma$$
by the invariance of the standard GAUSSian measure γ for the OU-semigroup. $\square$

ACHTUNG: The POINCARÉ and LOG-SOBOLEV inequalities have the following interesting corollaries:

(14) $\quad \mathrm{Var}_\gamma(P_t f) \leq e^{-2t} \cdot \mathrm{Var}_\gamma(f), \quad \mathrm{Ent}_\gamma(P_t f) \leq e^{-2t} \cdot \mathrm{Ent}_\gamma(f)$

for all $t \geq 0$ and $f \in \mathbb{L}^2(\gamma)$ (resp., $f \in \mathbb{L}\log\mathbb{L}(\gamma)$).

Indeed, from (10)''', (10)', (10) we obtain $\mathrm{Var}_{\gamma}(P_t f) = \underbrace{V^{P_t f}(0) - V^{P_t f}(\infty) = \int_{\mathbb{R}^n} \big(P_0(P_t f)\big)^2 d\gamma - \int_{\mathbb{R}^n} \big(P_\infty(P_t f)\big)^2 d\gamma} = V^f(t) - V^f(\infty)$,

thus

$$\frac{d}{dt} \mathrm{Var}_{\gamma}(P_t f) = -2 \int_{\mathbb{R}^n} |\nabla (P_t f)|^2 d\gamma \le -2 \cdot \mathrm{Var}_{\gamma}(P_t f),$$

and the first inequality in (14) follows from the GRONWALL inequality. Likewise, from (12)–(13) we get

$$\frac{d}{dt} \mathrm{Ent}_{\gamma}(P_t f) = - \int_{\mathbb{R}^n} \frac{|\nabla (P_t f)|^2}{P_t f} d\gamma \le -2 \cdot \mathrm{Ent}_{\gamma}(P_t f),$$

from which the second inequality in (14) follows. $\square$

<u>Remark:</u> In addition, we have the TALAGRAND Inequality discussed in class (cf. Exercise 6 in HOWK 10):

$$\underline{W_2(\mu, \gamma) \le \sqrt{2 \cdot H(\mu | \gamma)}} \qquad \text{for } \mu \in \mathcal{P}_2(\mathbb{R}^n).$$

Quadratic Wasserstein Distance

K-L Relative Entropy

$$\mathbb{E}^{\mu}\left[\log\left(\frac{d\mu}{d\gamma}\right)\right] = \int_{\mathbb{R}^n} f \log f \, d\gamma, \quad f \triangleq \frac{d\mu}{d\gamma}$$

Recall, from another Exercise, the PINSKER – CSISZÁR Inequality

$$\underline{\| \mu - \gamma \|_{TV} \le C \cdot H(\mu | \gamma)}$$

valid for arbitrary probability measures μ, γ.

L_iS - DIFFUSIONS

We fix now a "potential", meaning a function $\Psi : \mathbb{R}^n \to [0,\infty)$ of class $C^2(\mathbb{R}^n)$, and consider the corresponding LANGEVIN–SCHMOLUCHOWSKI diffusion

$$dX(t) = -\nabla\Psi(X(t))\,dt + \sqrt{2}\,dB(t)$$

with $B(\cdot)$ standard Brownian Motion in $\mathbb{R}^n$. Assuming

$$Z = \int_{\mathbb{R}^n} e^{-\Psi(x)}\,dx < \infty ,$$

this has

$$\gamma_\psi(dy) = \frac{1}{Z}\, e^{-\Psi(y)}\,dy$$

as its invariant distribution (just as the standard GAUSSian is the —"— of the OU-process when $\Psi(x) = |x|^2/2$).

The infinitesimal generator of this L_iS process is

(15) $$Lf = \Delta f - \langle \nabla\Psi, \nabla f \rangle ,$$

again by analogy with (1). And it is checked easily, that we have again the properties

(16)′ $$\int_{\mathbb{R}^n} Lf \cdot d\gamma_\Psi = 0$$

(16) $$\int_{\mathbb{R}^n} g \cdot Lf \cdot d\gamma_\Psi = \int_{\mathbb{R}^n} f \cdot Lg \cdot d\gamma_\Psi = -\int_{\mathbb{R}^n} \langle \nabla f, \nabla g \rangle\, d\gamma_\Psi$$

as in (5), (5)′.

9.a

We re-write this expression in terms of the positive operator $-L$ and the DIRICHLET Form

$$(16)'' \quad \underline{\mathcal{E}_\psi(f,g) \triangleq \int_{\mathbb{R}^n} \langle \nabla f, \nabla g \rangle \, d\gamma_\psi} = \int_{\mathbb{R}^n} g(-Lf)\, d\gamma_\psi = \int_{\mathbb{R}^n} f(-Lg)\, d\gamma_\psi \, .$$

We recall now the expression of Exercise 2 in HWK #10, which involves path integration on the space of continuous functions under WIENER Measure, and note that we do NOT have now an expression for the semigroup

$$\mathbb{R}^n \ni x \longmapsto (P_t f)(x) = \mathbb{E}^x\big(f(X(t))\big) \quad , \quad t \geq 0$$

as explicit and crisp as in the OU case.

So, in order to study this semigroup and obtain for it functional inequalities of the LOG-SOBOLEV or POINCARÉ type, we need to develop some new machinery.

To this end we introduce, with BAKRY-ÉMERY (1985), the Bilinear Form

(17) $$\Gamma(f,g) \triangleq \frac{1}{2}\Big[L(fg) - f\cdot L(g) - g\cdot L(f)\Big] = \langle \nabla f, \nabla g \rangle$$

Exercise

defined for functions f, g in a suitable set $A \subset \mathbb{L}^2(\gamma_\Psi)$ which is dense in $\mathbb{L}^2(\gamma_\Psi)$

(in the OU case, we can take $A = C_c^\infty(\mathbb{R}^n)$). We denote

(17)′ $$\Gamma(f) \equiv \Gamma(f,f) = \frac{1}{2}L(f^2) - f\cdot L(f) = |\nabla f|^2$$

(18) $$\Gamma_2(f) \triangleq \frac{1}{2}L(\Gamma(f)) - \Gamma(f, Lf) = \|D^2 f\|_{HS} + \langle \nabla f, D^2\Psi \cdot \nabla f \rangle$$

Exercise

Here and below, $D^2 f \; \left(D^2_{ij} f\right)_{1 \leq i,j \leq n}$ denotes the HESSIAN matrix of a function $f \in C^2(\mathbb{R}^n)$, and

$$\|D^2 f\|_{HS} \triangleq \sqrt{\sum_i \sum_j (D^2_{ij} f)^2}$$

is the HILBERT–SCHMIDT norm of this matrix.

PROPOSITION: BAKRY – ÉMERY (1985)

Suppose the potential Ψ satisfies $D^2\Psi(x) \geq \rho \cdot Id$

for all $x \in \mathbb{R}^n$; i.e.,

(19) $$\langle y, D^2\Psi(x)\, y\rangle \geq \rho\, |y|^2, \quad \forall\, y \in \mathbb{R}^n$$

for some $\rho \in \mathbb{R}$.

Then

(20) $$\Gamma_2(f) \geq \rho \cdot \Gamma(f), \quad \forall\, f \in \mathcal{A}.$$

THEOREM: Suppose the condition (19) is satisfied for some $\rho \in (0,\infty)$. Then we have the POINCARÉ and LOG–SOBOLEV inequalities

(21) $$\mathrm{Var}_{\gamma_\psi}(f) \leq \frac{1}{\rho}\int_{\mathbb{R}^n} |\nabla f|^2 \, d\gamma_\psi, \qquad \mathrm{Ent}_{\gamma_\psi}(f) \leq \frac{1}{2\rho}\int_{\mathbb{R}^n} \frac{|\nabla f|^2}{f}\, d\gamma_\psi$$

as well as the exponential decay of the variance and relative entropy

(22) $$\mathrm{Var}_{\gamma_\psi}(P_t f) \leq e^{-\frac{2t}{\rho}}\, \mathrm{Var}_{\gamma_\psi}(f), \qquad \mathrm{Ent}_{\gamma_\psi}(P_t f) \leq e^{-\frac{2t}{\rho}}\, \mathrm{Ent}_{\gamma_\psi}(f)$$

for $f \in \mathcal{A}$ (resp., $f > 0$, $f \in \mathcal{A}$).

<u>Remark:</u> More generally, we define in the spirit of (18) the second-order bilinear form

(18)′ $$\Gamma_2(f,g) = \frac{1}{2}\Big[L\big(\Gamma(f,g)\big) - \Gamma(f, Lg) - \Gamma(g, Lf)\Big], \quad (f,g)\in \mathcal{A}\times\mathcal{A}.$$

As regards the claim of (20), it amounts to

$$\| D^2 f\|_{HS} + \langle \nabla f, D^2\Psi \cdot \nabla f\rangle \geqslant \rho \cdot |\nabla f|^2$$

on account of the "Exercises" in (17), (17)′, (18); and is obviously satisfied under the condition (19).

<u>Remark:</u> The adjoint L^* of the generator L in (15) is

(15)* $$L^* g = \Delta g + \mathrm{div}\,(g \cdot \nabla \Psi)$$

This is clear via integration by parts

$$\int_{\mathbb{R}^n} Lf \cdot g \cdot d\gamma_\psi = \int_{\mathbb{R}^n} \big(\Delta f - \langle \nabla f, \nabla \Psi\rangle\big)\, g\, d\gamma_\psi = \int_{\mathbb{R}^n} f\big(\Delta g + \mathrm{div}(g\cdot\nabla\Psi)\big)\, d\gamma_\psi$$

and leads to the FOKKER–PLANCK–KOLMOGOROV equation

$$\frac{\partial}{\partial t} p_t = \Delta p_t + \mathrm{div}\,\big(p_t \cdot \nabla \Psi\big), \qquad t>0$$

for the probability density function $p_t(\cdot)$ of $X(t)$.

<u>Remark:</u> Both inequalities in (21) can be written in terms of the DIRICHLET Energy

$$\mathcal{E}_{\psi}(f,f) \triangleq \int_{\mathbb{R}^n} |\nabla f|^2 \, d\gamma_{\psi}$$

as

(21)' $$\operatorname{Var}_{\gamma_\psi}(f) \leq \frac{1}{\rho}\, \mathcal{E}_{\psi}(f,f)\,, \quad \operatorname{Ent}_{\gamma_\psi}(f^2) \leq \frac{2}{\rho}\, \mathcal{E}_{\psi}(f,f)\,.$$

<u>Remark:</u> Once again, with the KL Relative Entropy

$$H(\mu|\gamma_\psi) = \mathbb{E}^{\mu}\left[\log\left(\frac{d\mu}{d\gamma_\psi}\right)\right] = \int_{\mathbb{R}^n} f \log f \, d\gamma_\psi\,, \quad f \triangleq \frac{d\mu}{d\gamma_\psi}$$

and the FISHER Information

$$I_{\gamma_\psi}(\mu) \triangleq \int_{\mathbb{R}} \frac{|\nabla f|^2}{f}\, d\gamma_\psi\,,$$

we have the OTTO-VILLANI (2000) <u>HWI Inequality</u>

(23) $$H(\mu|\gamma_\psi) - H(\nu|\gamma_\psi) \leq W_2(\mu,\nu)\cdot\sqrt{I_{\gamma_\psi}(\mu)} - \frac{\rho}{2}\cdot W_2^2(\mu,\nu)$$

for any two probability measures μ, ν in $\mathcal{P}_2(\mathbb{R}^n)$. This implies the <u>TALAGRAND Inequality</u>

(24) $$W_2(\mu,\gamma_\psi) \leq \sqrt{\frac{2}{\rho}\cdot H(\mu|\gamma_\psi)}$$

and the <u>Modified LOG-SOBOLEV</u>

(25) $$H(\mu|\gamma_\psi) \leq \frac{1}{2\rho}\, I_{\gamma_\psi}(\mu)\,.$$

Remark: Once again, the LOG-SOBOLEV inequality in (21) can be cast, always under the assumption (19), in the equivalent form

(21)′
$$\frac{\rho}{2}\,\mathrm{Ent}_{\gamma_\psi}(f^2) \leq \int_{\mathbb{R}^n} |\nabla f|^2 \, d\gamma_\psi \,.$$

GROSS (1975) proved that this inequality implies (in fact, is equivalent to) the following property of hypercontractivity.

DEFINITION: The LS-semigroup $(P_t f)_{t \geq 0}$ is called hypercontractive if, for some (then all) $1 < p < q < \infty$

$$\| P_t f \|_q \leq \| f \|_p \quad , \quad \forall\, f \in \mathbb{L}^p(\gamma_\psi)$$

holds for all $t > 0$ large enough, namely, $t > \frac{1}{2\rho} \log\left(\frac{q-1}{p-1}\right)$.

An interesting proof of this result in the special case of the OU-semigroup, based on Stochastic Analysis, appears on pp. 367–369 of IKEDA–WATANABE (1988).

Remark: The TALAGRAND Inequality (24) is implied by the LOG-SOBOLEV Inequality in (21); see p. 126, Theorem 6.6 of LEDOUX's book.

CONCENTRATION OF MEASURE

It turns out that a LOG-SOBOLEV inequality of the form

(26) $$\mathrm{Ent}_\mu(f^2) \le 2C \int_{\mathbb{R}^n} |\nabla f|^2 \, d\mu$$

for every sufficiently smooth function $f: \mathbb{R}^n \to \mathbb{R}$ and relative to a given probability measure μ on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$, implies the concentration of measure inequality

(27) $$\mu\left(\left\{F - \int_{\mathbb{R}} F\,d\mu \ge r\right\}\right) \le e^{-r^2/2C}, \quad r > 0$$

for every integrable and Lip(1)-function $F: \mathbb{R}^n \to \mathbb{R}$.

(Here, we can replace $\mathbb{R}^n$ by an arbitrary metric space equipped with the BOREL σ-algebra generated by its open sets.)

ACHTUNG: The right-hand side in (27) does NOT involve n.

Proof: Consider an integrable, LIPSCHITZ-continuous function $F: \mathbb{R}^n \to \mathbb{R}$ with $\int_{\mathbb{R}^n} F\,d\mu = 0$. Assuming, as we may, that F is smooth enough, we may take $|\nabla F| \le \|F\|_{Lip}$ everywhere. We apply then (26) to the function

(28) $$f = e^{\frac{\lambda}{2}F - \frac{\lambda^2}{4}C\|F\|^2_{Lip}}$$

for $\lambda \in \mathbb{R}$, for which

$$\int_{\mathbb{R}^n} |\nabla f|^2 \, d\mu = \frac{\lambda^2}{4} \int_{\mathbb{R}^n} |\nabla F|^2 e^{\lambda F - \frac{C}{2}\lambda^2 \|F\|^2_{Lip}} \, d\mu$$

$$\le \frac{\lambda^2}{4} \|F\|^2_{Lip} \int_{\mathbb{R}^n} e^{\lambda F - \frac{C}{2}\lambda^2\|F\|^2_{Lip}} \, d\mu .$$

With

(29) $$K(\lambda) \triangleq \int_{\mathbb{R}^n} e^{\lambda F - \frac{c}{2}\lambda^2 \|F\|_{Lip}^2}\, d\mu$$

and the choice (28), the LSI inequality (26) gives

$$\int_{\mathbb{R}^n} \left(\lambda F - \frac{c}{2}\lambda^2 \|F\|_{Lip}^2\right) e^{\lambda F - c\lambda^2 \|F\|_{Lip}^2/2}\, d\mu - K(\lambda) . \log K(\lambda)$$
$$\leq \frac{c}{2}\lambda^2 \|F\|_{Lip}^2 \cdot K(\lambda).$$

But this, amounts to

(30) $$\lambda K'(\lambda) \leq K(\lambda) . \log K(\lambda) \quad , \quad \lambda \in \mathbb{R} .$$

Now with $H(\lambda) \triangleq \frac{1}{\lambda} \log K(\lambda)$, $\lambda \in \mathbb{R}$ (thus, by L'HOPITAL, $H(0) = \frac{K'(0)}{K(0)} = \int_{\mathbb{R}} F\, d\mu = 0$), the inequality (30) amounts to $H'(\lambda) \leq 0$ for $\lambda \in \mathbb{R}$. Thus $H(\cdot)$ is non-increasing, which means $e^{\lambda . H(\lambda)} = K(\lambda) \leq 1$, $\forall\, \lambda \geq 0$. In other words

(31) $$\int_{\mathbb{R}^n} e^{\lambda F}\, d\mu \leq e^{\frac{c}{2}\lambda^2 \|F\|_{Lip}^2}$$

from (29), for all $\lambda \geq 0$. And replacing F by a smooth convolution, we see that (31) holds for every Lipschitz function with zero mean.

If, in addition, $\|F\|_{Lip} \leq 1$, then

$$\mu(\{F \geq r\}) \leq e^{-\lambda r} \int_{\mathbb{R}^n} e^{\lambda F}\, d\mu \leq e^{c\lambda^2/2 - \lambda r} \quad , \quad r \geq 0 .$$

Optimizing over $\lambda \geq 0$, we obtain $\mu(\{F \geq r\}) \leq e^{-r^2/2c}$, and the claim (27) follows readily. $\square$